| QUALIFICATION |
| REQUIREMENTS |
| REMOVED |

MIL-M-38510/158A
23 April 1984
SUPERSEDING
MIL-M-38510/158(USAF)
8 December 1975

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, DECODERS,
MONOLITHIC SILICON

[INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, decoder microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as specified in MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

| Device type | Circuit |
|---|---|
| 01 | 1 of 4 decoder |
| 02 | 7 segment decoder/driver |

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

| Outline letter | Case outline (see MIL-M-38510, appendix C) |
|---|---|
| E | D-2 (16-pin, 1/4" x 7/8"), dual-in-line package |
| F | F-5 (16-pin, 1/4" x 3/8"), flat package |

1.2.4 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range | -0.5 V dc to +7.0 V dc |
| Input voltage range | -1.5 V dc at -12 mA to +5.5 V dc |
| Storage temperature range | -65°C to +150°C |
| Maximum power dissipation ($P_D$) 1/ | |
| Device type 01 | 275 mWdc |
| Device type 02 | 400 mWdc |
| Lead temperature (soldering, 10 seconds) | +300°C |
| Thermal resistance, junction-to-case ($\theta_{JC}$): | |
| Case E | 50°C/W |
| Case F | 70°C/W |
| Junction temperature ($T_J$) | +175°C |

1/ Must withstand the added $P_D$ due to short circuit test (e.g., $I_{OS}$) at one output for 5 seconds duration.

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |
|---|

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1.2.5 Recommended operating conditions.

| | |
|---|---|
| Supply voltage ($V_{CC}$) | 4.5 V dc minimum to 5.5 V dc maximum |
| Minimum high-level input voltage ($V_{IH}$) | 2.0 V dc |
| Maximum low-level input voltage ($V_{IL}$) | 0.8 V dc |
| Sink-current capability by device type | |
| 01 | 16 mA |
| 02, outputs a - g | 20 mA |
| 02, RBO node | 2.4 mA |
| Case operating temperature range ($T_C$) | -55°C to +125°C |

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 2.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.5 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.



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TABLE I. Electrical performance characteristics.

| Tests | Symbol | Conditions 1/ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| High-level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $I_{OH}$ = -0.8 mA<br>$V_{IN}$ = 0.8 V and 2.0 V | 01 | 2.4 | | V |
| Low-level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $I_{OL}$ = 16 mA<br>$V_{IN}$ = 0.8 and 2.0 V | 01 | | 0.4 | V |
| Low-level output voltage | $V_{OL1}$ | $V_{CC}$ = 4.5 V, $I_{OL}$ = 2.4 mA 4/<br>$V_{IN}$ = 0.8 and 2.0 V | 02 | | 0.4 | V |
| Low-level output voltage | $V_{OL2}$ | $V_{CC}$ = 5.5 V, $I_{OL}$ = 3.1 mA 4/<br>$V_{IN}$ = 0.8 and 2.0 V | 02 | | 0.4 | V |
| Low-level output voltage | $V_{OL3}$ | $V_{CC}$ = 4.5 V, $I_{OL}$ = 20 mA<br>$\overline{LT}$ = 0 V | 02 | | 0.4 | V |
| High-level output voltage (RBO output only) | $V_{OH}$ | $V_{CC}$ = 4.5 V, $I_{OH}$ = -70 µA<br>$V_{IN}$ = 0.8 V and 2.0 V | 02 | 3.0 | | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V, $I_{IN}$ = -12 mA | 01 | | -1.5 | V |
| Low-level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | 01 | -0.7 | -1.6 | mA |
| Low-level input current (data inputs) | $I_{IL1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | 02 | -0.5 | -1.4 | mA |
| Low-level input current ($\overline{LT}$ input) | $I_{IL2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | 02 | -2.8 | -6.4 | mA |
| Low-level input current ($\overline{RBI}$ input) | $I_{IL3}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | 02 | -0.3 | -0.75 | mA |
| High-level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V 2/<br>(except $\overline{LT}$ input on 02) | 01, 02 | 0 | 40 | µA |
| High-level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V 2/<br>(except $\overline{LT}$ input on 02) | 01, 02 | 0 | 100 | µA |
| High-level input current | $I_{IH3}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V 2/<br>(at $\overline{LT}$ only) | 02 | 0 | 200 | µA |
| High-level input current | $I_{IH4}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V 2/<br>(at $\overline{LT}$ only) | 02 | 0 | 500 | µA |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Tests | Symbol | Conditions 1/ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V 2/ 3/ | 01 | -20 | -70 | mA |
| Short circuit output current ($\overline{RBO}$ output only) | $I_{OS}$ | $V_{CC}$ = 4.5 V 4/ | 02 | -0.64 | | mA |
| Short circuit output current ($\overline{RBO}$ output only) | $I_{OS}$ | $V_{CC}$ = 5.5 V 4/ | 02 | | -6.8 | mA |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0 V | 01 | | 50 | mA |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.0 V | 02 | | 66 | mA |
| Output latching voltage | $B_{VO}$ | $V_{CC}$ = 5.0 V, $I_{OUT}$ = 10 mA | 02 | 30 | | V |
| High-level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V 2/ (except $\overline{LT}$ input on 02) | 01, 02 | 0 | 100 | µA |
| High-level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V 2/ (at $\overline{LT}$ only) | 02 | 0 | 500 | µA |
| Propagation delay time from $\overline{E}$ to $\overline{O}$, $\overline{I}$, 2, 3 | $t_{PLH}$ | $V_{CC}$ = 5.0 V<br>$R_L$ = 390Ω ±5%<br>$C_L$ = 50 pF minimum<br>See figure 6 | 01 | 6 | 39 | ns |
| | $t_{PHL}$ | | 01 | 12 | 35 | ns |
| Propagation delay time from A0 to $\overline{O}$, $\overline{2}$<br>A1 to $\overline{O}$, $\overline{I}$ | $t_{PLH}$ | | 01 | 6 | 39 | ns |
| | $t_{PHL}$ | | 01 | 6 | 35 | ns |
| Propagation delay time from A0 to $\overline{I}$, $\overline{3}$<br>A1 to $\overline{2}$, $\overline{3}$ | $t_{PLH}$ | | 01 | 9 | 50 | ns |
| | $t_{PHL}$ | | 01 | 9 | 41 | ns |
| Propagation delay time from A0, A1, A2, A3, $\overline{LT}$ to $\overline{a}$, $\overline{b}$, $\overline{c}$, $\overline{d}$, $\overline{e}$, $\overline{f}$, $\overline{g}$ | $t_{PLH}$ | $V_{CC}$ = 5.0 V<br>$R_L$ = 240Ω ±5%<br>$C_L$ = 50 pF minimum<br>$R_L$ = 2,000Ω ±5%<br>See figure 7 | 02 | 30 | 1,100 | ns |
| | $t_{PHL}$ | | 02 | 50 | 1,200 | ns |
| Propagation delay time from A0, A1, A2, A3 to $\overline{RBO}$ | $t_{PLH}$ | | 02 | 60 | 1,100 | ns |
| | $t_{PHL}$ | | 02 | 10 | 500 | ns |
| Propagation delay time from $\overline{RBI}$ to $\overline{a}$, $\overline{b}$, $\overline{c}$, $\overline{d}$, $\overline{e}$, $\overline{f}$, $\overline{g}$ | $t_{PLH}$ | | 02 | 30 | 1,200 | ns |
| | $t_{PHL}$ | | 02 | 30 | 1,300 | ns |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Tests | Symbol | Conditions 1/ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Propagation delay time from $\overline{RBI}$ to $\overline{RBO}$ | $t_{PLH}$ | $V_{CC}$ = 5.0 V<br>RL = 240Ω ±5%<br>CL = 50 pF minimum<br>RL = 2,000Ω ±5%<br>See figure 7 | 02 | 40 | 700 | ns |
| | $t_{PHL}$ | | 02 | 10 | 400 | ns |
| Propagation delay time from $\overline{LT}$ to $\overline{RBO}$ | $t_{PLH}$ | | 02 | 30 | 700 | ns |
| | $t_{PHL}$ | | 02 | 10 | 80 | ns |
| Propagation delay time from $\overline{RBO}$ to $\overline{a}$, $\overline{b}$, $\overline{c}$, $\overline{d}$, $\overline{e}$, $\overline{f}$, $\overline{g}$ | $t_{PLH}$ | | 02 | 30 | 700 | ns |
| | $t_{PHL}$ | | 02 | 30 | 600 | ns |

1/ Complete terminal conditions shall be as specified in table III.

2/ All unspecified inputs grounded.

3/ Not more than one output should be shorted at one time.

4/ $\overline{RBO}$ node only.



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3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line; not necessarily the line producing the device type described herein.

TABLE II. Electrical test requirements.

| MIL-STD-883 test requirements | Subgroups (see table III) Class B devices |
|---|---|
| Interim electrical parameters (pre burn-in) (method 5004) | 1 |
| Final electrical test parameters (method 5004) | 1*, 2, 3, 7, 9 |
| Group A test requirements (method 5005) | 1, 2, 3, 7, 9, 10, 11 |
| Groups C and D end-point electrical parameters (method 5005) | 1, 2, 3 |
| Additional electrical subgroups for group C periodic inspections | None |

*PDA applies to subgroup 1 (see 4.2c).

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) $T_A$ = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA.



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The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6, of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) $T_A$ = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be as specified in MIL-M-38510.

6. NOTES

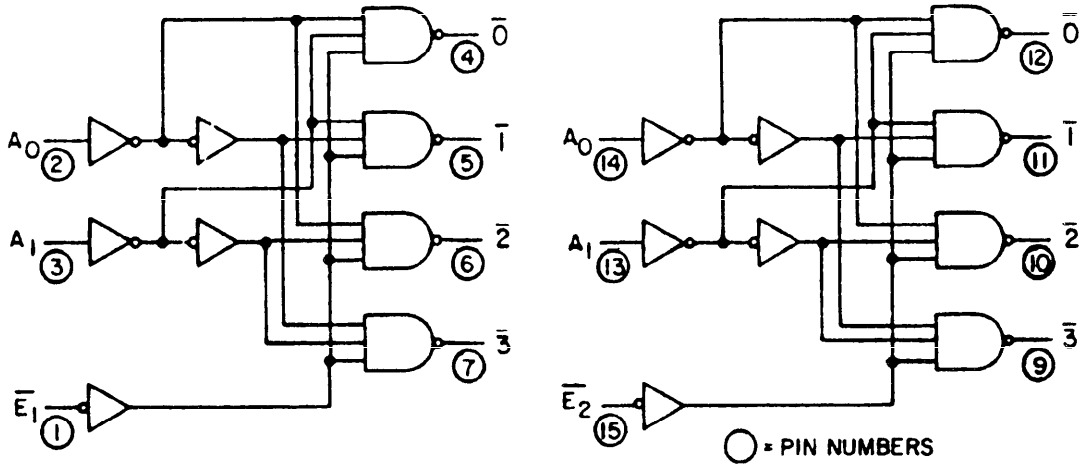
6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logisitic support of existing equipment.



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Device type 01




Device type 02

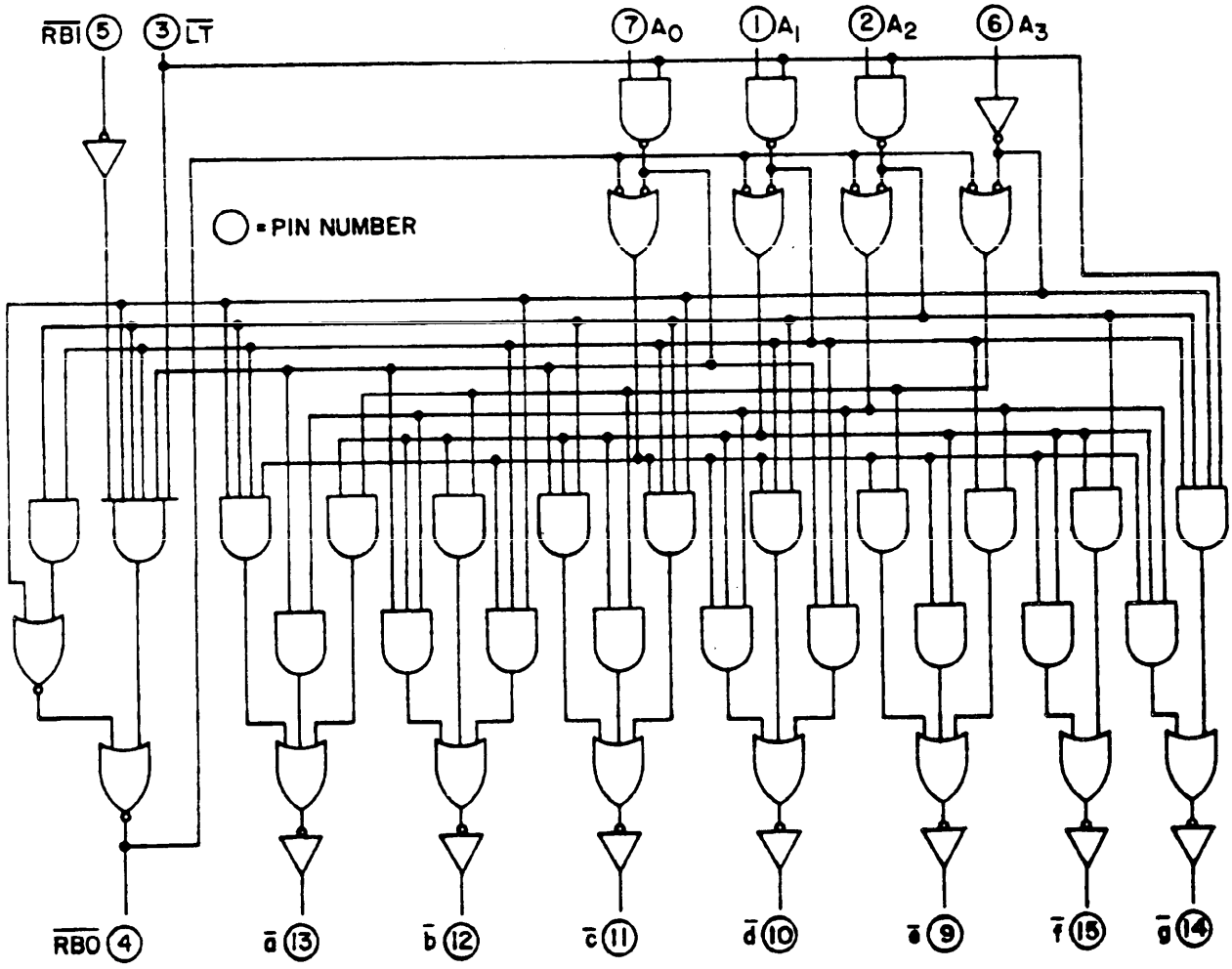


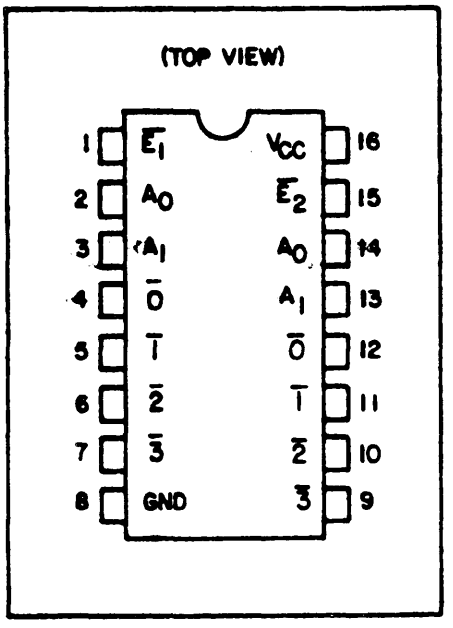

FIGURE 1. Logic diagrams.



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Device type 01

CASES E AND F




Device type 02

CASES E AND F

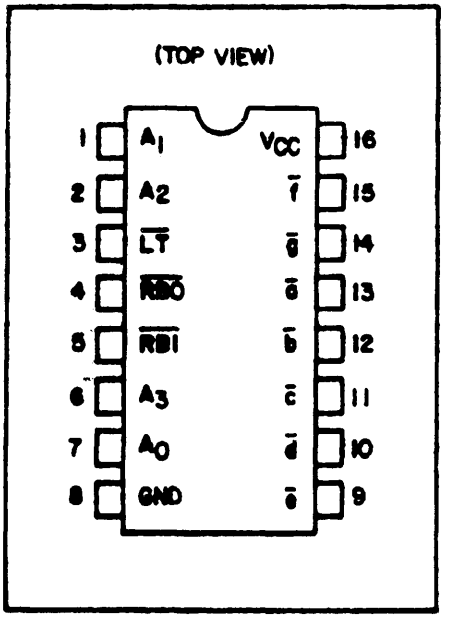



FIGURE 2. Terminal connections.



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**Device type 01**

**DECODER 1 & 2**

| $\overline{E}$ | $A_0$ | $A_1$ | $\overline{0}$ | $\overline{1}$ | $\overline{2}$ | $\overline{3}$ |
|---|---|---|---|---|---|---|
| L | L | L | L | H | H | H |
| L | H | L | H | L | H | H |
| L | L | H | H | H | L | H |
| L | H | H | H | H | H | L |
| H | X | X | H | H | H | H |

H = HIGH voltage level
L = LOW voltage level
X = Irrelevant

**Device type 02**

| $\overline{LT}$ | $\overline{RBI}$ | $A_0$ | $A_1$ | $A_2$ | $A_3$ | $\overline{a}$ | $\overline{b}$ | $\overline{c}$ | $\overline{d}$ | $\overline{e}$ | $\overline{f}$ | $\overline{g}$ | $\overline{RBO}$ | DECIMAL OR FUNCTION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L | X | X | X | X | X | L | L | L | L | L | L | L | H | |
| H | L | L | L | L | L | H | H | H | H | H | H | H | L | 0 |
| H | H | L | L | L | L | L | L | L | L | L | L | H | H | 0 |
| \| | X | H | L | L | L | H | H | H | H | L | L | H | H | 1 |
| \| | \| | L | H | L | L | L | L | H | L | L | H | L | H | 2 |
| \| | \| | H | H | L | L | L | L | L | L | H | H | L | H | 3 |
| \| | \| | L | L | H | L | H | L | L | H | H | L | L | H | 4 |
| \| | \| | H | L | H | L | L | H | L | L | H | L | L | H | 5 |
| \| | \| | L | H | H | L | H | H | L | L | L | L | L | H | 6 |
| \| | \| | H | H | H | L | L | L | L | H | H | H | H | H | 7 |
| \| | \| | L | L | L | H | L | L | L | L | L | L | L | H | 8 |
| \| | \| | H | L | L | H | L | L | L | H | H | L | L | H | 9 |
| \| | \| | L | H | L | H | H | H | H | H | H | H | H | L | 10 |
| \| | \| | H | H | L | H | H | H | H | H | H | H | H | L | 11 |
| \| | \| | L | L | H | H | H | H | H | H | H | H | H | L | 12 |
| \| | \| | H | L | H | H | H | H | H | H | H | H | H | L | 13 |
| \| | \| | L | H | H | H | H | H | H | H | H | H | H | L | 14 |
| H | X | H | H | H | H | H | H | H | H | H | H | H | L | 15 |

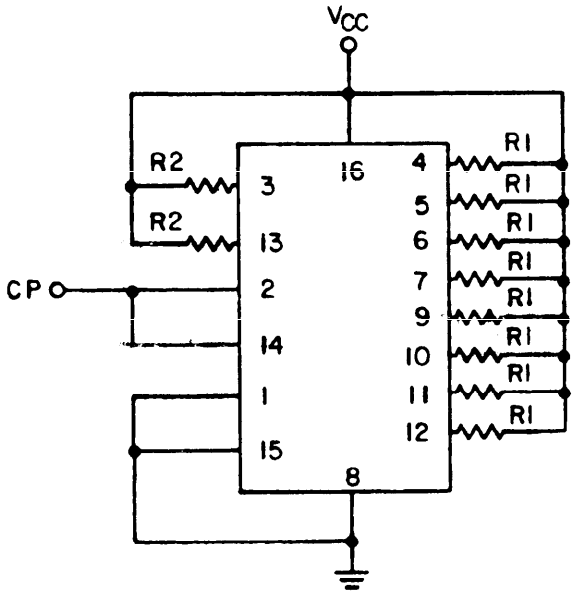
H = HIGH voltage level
L = LOW voltage level
X = Irrelevant

FIGURE 3. Truth tables.



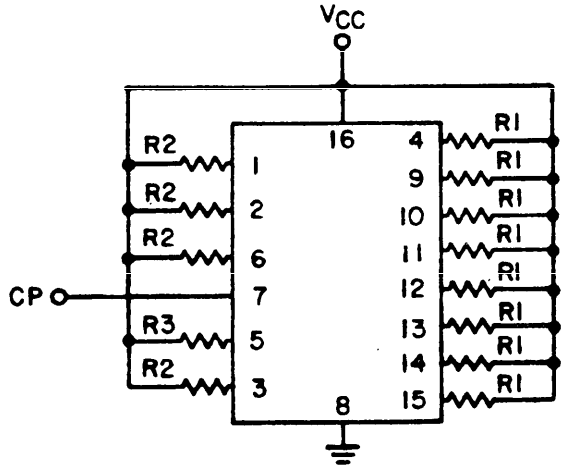
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Device type 01




NOTES:
1. CP = 3 volts, frequency = 1 MHz, and 50% duty cycle.
2. $V_{CC}$ = 5.5 V.
3. R1 = 318Ω, R2 = 4 kΩ.

Device type 02

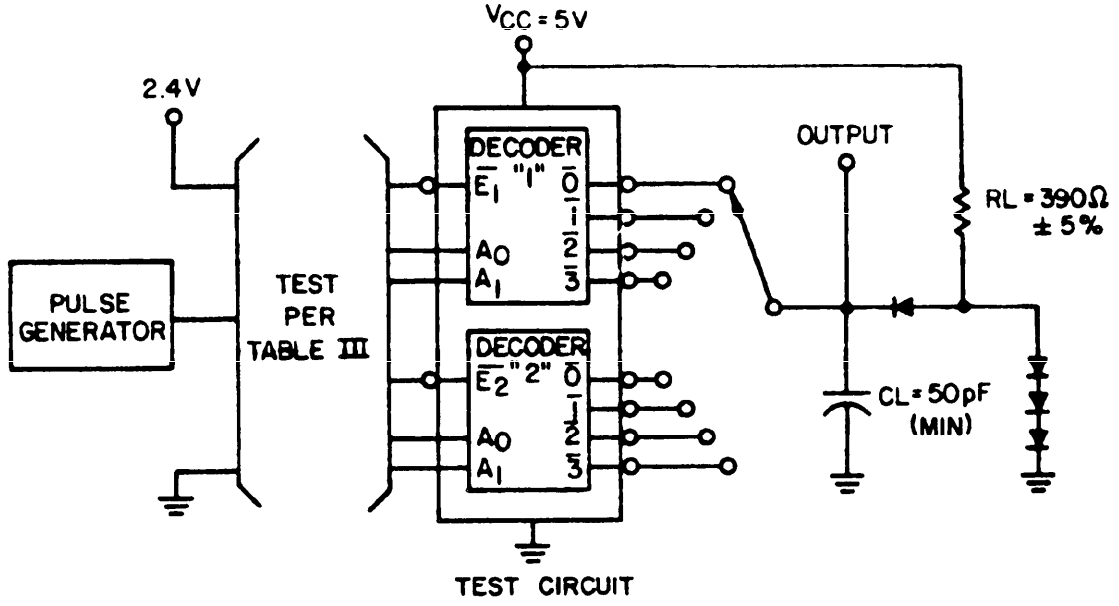



NOTES:
1. CP = 3 volts, frequency = 1 MHz, and 50% duty cycle.
2. VCC = 5.5 V.
3. R1 = 255Ω, R2 = 2.6 kΩ, R3 = 5.2 kΩ.

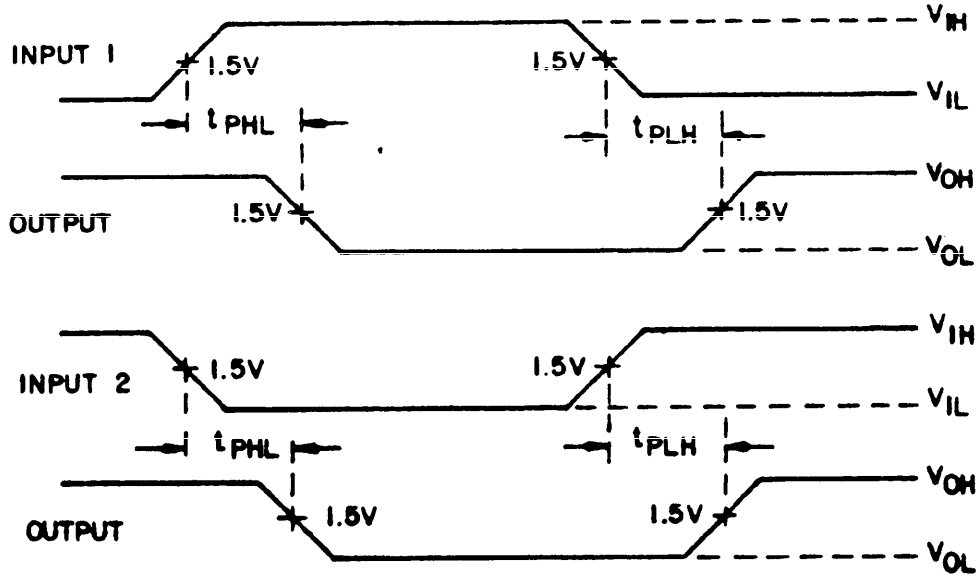
FIGURE 4. Burn-in and life test circuit.



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TEST CIRCUIT

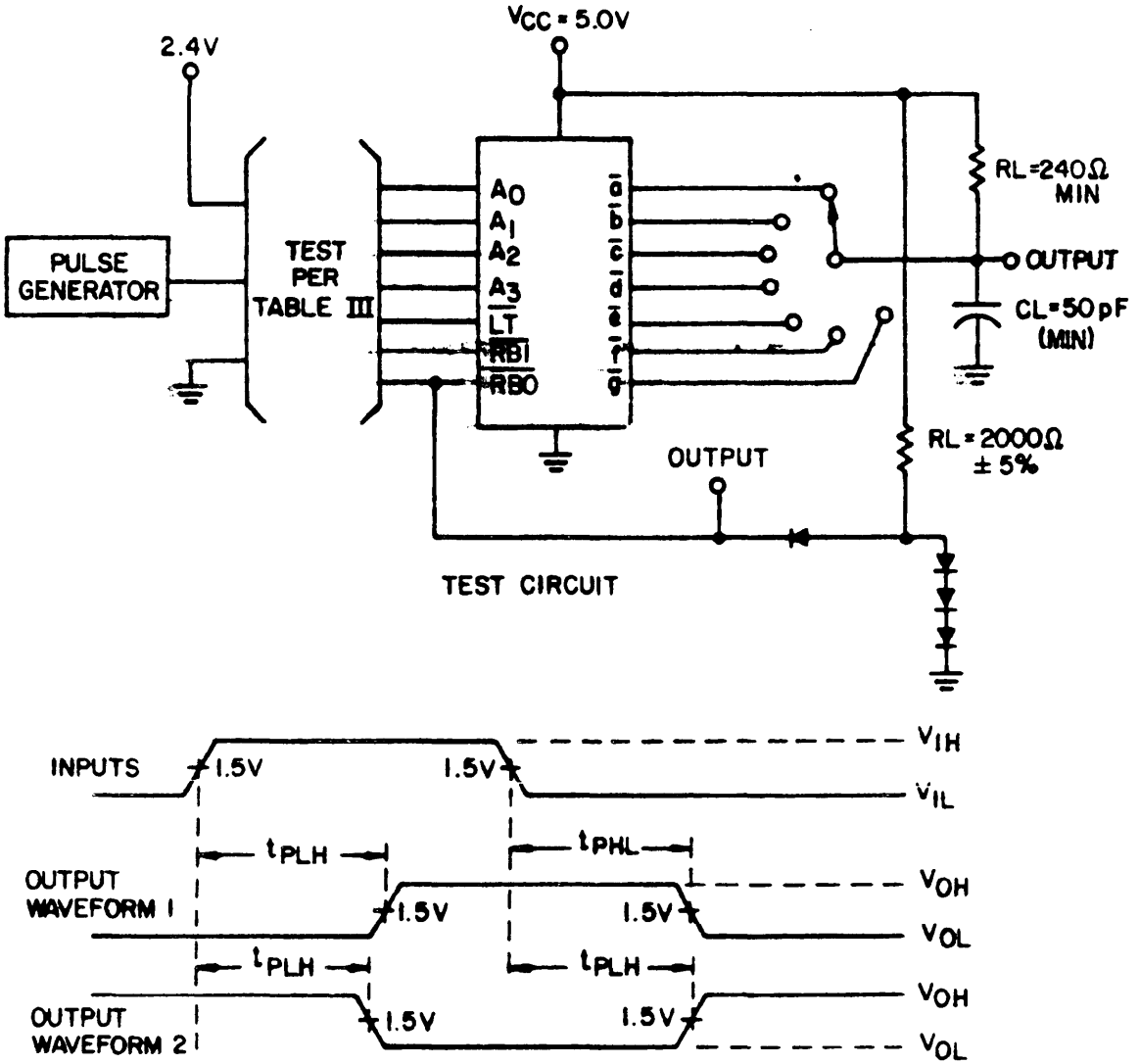



NOTES:
1. The pulse generator has the following characteristics: Vgen = 3.0 V min. $t_{TLH}$ (0.7 V to 2.7 V) and $t_{THL}$ (2.7 V to 0.7 V) 10 ns, PRR = 1 MHz, and min. duty cycle 50%.
2. CL includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Input 1 and input 2 are used to test each output with each A input. Input 2 is used to test each output with each E input.

FIGURE 5. Switching time for device type 01.



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NOTES:
1. The pulse generator has the following characteristics: $V_{gen}$ = 3.0 V min. $t_{TLH}$ (0.7 V to 2.7 V) and $t_{THL}$ (2.7 V to 0.7 V) 10 ns, PRR = 1 MHz, and min. duty cycle 50%.
2. $C_L$ includes probe and jig capacitance.
3. Input - output waveform combination in accordance with the truth tables.
4. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching time for device type 02.



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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high $\geq$ 2.0 V; or low $\leq$ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 $\overline{E1}$ | 2 A0 | 3 A1 | 4 $\overline{0}$ | 5 $\overline{1}$ | 6 $\overline{2}$ | 7 $\overline{3}$ | 8 GND | 9 $\overline{3}$ | 10 $\overline{2}$ | 11 $\overline{1}$ | 12 $\overline{0}$ | 13 A1 | 14 A0 | 15 $\overline{E2}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1<br>$T_C$ = 25°C | $V_{OH}$ | 3007 | 1<br>2<br>3<br>4<br>5<br>6<br>7<br>8 | 2.0 V | GND<br>4.5 V<br>GND<br>4.5 V | GND<br>GND<br>4.5 V<br>4.5 V | -.8 mA | -.8 mA | -.8 mA | -.8 mA | GND | -.8 mA | -.8 mA | -.8 mA | -.8 mA | GND<br>GND<br>4.5 V<br>4.5 V | GND<br>4.5 V<br>GND<br>4.5 V | 2.0 V | 4.5 V | $\overline{0}$<br>1<br>2<br>3<br>$\overline{0}$<br>1<br>2<br>3 | 2.4 | | V |
| | $V_{OL}$ | 3006 | 9<br>10<br>11<br>12<br>13<br>14<br>15<br>16 | 0.8 V<br>GND<br>GND<br>GND | 0.8 V<br>2.0 V<br>GND<br>4.5 V | 0.8 V<br>GND<br>2.0 V<br>4.5 V | 16 mA | 16 mA | 16 mA | 16 mA | | 16 mA | 16 mA | 16 mA | 16 mA | 0.8 V<br>GND<br>2.0 V<br>4.5 V | 0.8 V<br>2.0 V<br>GND<br>4.5 V | 0.8 V<br>GND<br>GND<br>GND | | $\overline{0}$<br>1<br>2<br>3<br>$\overline{0}$<br>1<br>2<br>3 | | 0.4 | |
| | $V_{IC}$ | | 17<br>18<br>19<br>20<br>21<br>22 | -12 mA | -12 mA | -12 mA | | | | | | | | | | -12 mA | -12 mA | -12 mA | | $\overline{E1}$<br>A0<br>A1<br>$\overline{E2}$<br>A0<br>A1 | | -1.5 | |
| | $I_{IH1}$ | 3010 | 23<br>24<br>25<br>26<br>27<br>28 | 2.4 V | 2.4 V | 2.4 V | | | | | | | | | | 2.4 V | 2.4 V | 2.4 V | 5.5 V | $\overline{E1}$<br>A0<br>A1<br>$\overline{E2}$<br>A0<br>A1 | 0 | 40 | μA |
| | $I_{IH2}$ | | 29<br>30<br>31<br>32<br>33<br>34 | 5.5 V | 5.5 V | 5.5 V | | | | | | | | | | 5.5 V | 5.5 V | 5.5 V | | $\overline{E1}$<br>A0<br>A1<br>$\overline{E2}$<br>A0<br>A1 | | 100 | |
| | $I_{IL}$ | 3009 | 35<br>36<br>37<br>38<br>39<br>40 | 0.4 V | 0.4 V | 0.4 V | | | | | | | | | | 0.4 V | 0.4 V | 0.4 V | | $\overline{E1}$<br>A0<br>A1<br>$\overline{E2}$<br>A0<br>A1 | -0.7 | -1.6 | mA |
| | $I_{OS}$ | 3011 | 41<br>42<br>43<br>44<br>45<br>46<br>47<br>48 | GND | 4.5 V<br>GND<br>GND<br>4.5 V | GND | GND | GND | GND | GND | | GND | GND | GND | GND | GND | 4.5 V<br>GND<br>GND<br>4.5 V | GND | | $\overline{0}$<br>1<br>2<br>3<br>$\overline{0}$<br>1<br>2<br>3 | -20 | -70 | |
| | $I_{CC}$ | 3005 | 49 | GND | GND | GND | | | | | | | | | | | GND | | | $V_{CC}$ | | 50 | |

See notes at end of device type 01.





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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 $\overline{E1}$ | 2 A0 | 3 A1 | 4 $\overline{0}$ | 5 $\overline{1}$ | 6 $\overline{2}$ | 7 $\overline{3}$ | 8 GND | 9 $\overline{3}$ | 10 $\overline{2}$ | 11 $\overline{1}$ | 12 $\overline{0}$ | 13 A1 | 14 A0 | 15 $\overline{E2}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_A$ = 125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_A$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$ = 25°C | Truth table test | | 50<br>51<br>52<br>53<br>54 | GND<br>5.0 V | GND<br>5.0 V<br>GND<br>5.0 V<br>X | GND<br>GND<br>5.0 V<br>5.0 V<br>X | L<br>H | H<br>L<br>H<br>H<br>H | H<br>H<br>L<br>H<br>H | H<br>H<br>H<br>L<br>H | GND | H<br>H<br>H<br>L<br>H | H<br>H<br>L<br>H<br>H | H<br>L<br>H<br>H<br>H | L<br>H | GND<br>GND<br>5.0 V<br>5.0 V<br>X | GND<br>5.0 V<br>GND<br>5.0 V<br>X | GND<br>5.0 V | 5.0 V | | See notes 1 and 2 | | |
| 8 | Truth table test | | 55 thru 59 | Same tests as subgroup 7 except $T_C$ = 125°C. | | | | | | | | | | | | | | | | | | | |
| | Truth table test | | 60 thru 64 | Same tests as subgroup 7 except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$ = 25°C | $t_{PHL}$ | 3003 (Fig 5) | 65<br>66<br>67<br>68 | IN | GND<br>5.0 V<br>GND<br>5.0 V | GND<br>GND<br>5.0 V<br>5.0 V | OUT | OUT | OUT | OUT | GND | | | | | | | | 5.0 V | $\overline{E1}$ - 0<br>$\overline{E1}$ - 1<br>$\overline{E1}$ - 2<br>$\overline{E1}$ - 3 | 6 | 28 | ns |
| | $t_{PLH}$ | | 69<br>70<br>71<br>72 | | GND<br>5.0 V<br>GND<br>5.0 V | GND<br>GND<br>5.0 V<br>5.0 V | OUT | OUT | OUT | OUT | | | | | | | | | | $\overline{E1}$ - 0<br>$\overline{E1}$ - 1<br>$\overline{E1}$ - 2<br>$\overline{E1}$ - 3 | | 25 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 73<br>74 | GND | IN | GND<br>5.0 V | | OUT | OUT | | | | | | | | | | | A0 - 1<br>A0 - 2 | 9<br>6 | 33<br>28 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 75<br>76 | | | GND<br>5.0 V | | OUT | OUT | | | | | | | | | | | A0 - 1<br>A0 - 2 | 9<br>6 | 33<br>25 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 77<br>78 | | GND<br>5.0 V | IN | OUT | | | OUT | | | | | | | | | | A1 - 0<br>A1 - 3 | 6<br>9 | 28<br>33 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 79<br>80 | | GND<br>5.0 V | | OUT | | | OUT | | | | | | | | | | A1 - 0<br>A1 - 3 | 6<br>9 | 25<br>33 | |
| | $t_{PHL}$ | | 81<br>82<br>83<br>84 | | | | | | | | | OUT | OUT | OUT | OUT | GND<br>GND<br>5.0 V<br>5.0 V | GND<br>5.0 V<br>GND<br>5.0 V | IN | | $\overline{E2}$ - 0<br>$\overline{E2}$ - 1<br>$\overline{E2}$ - 2<br>$\overline{E2}$ - 3 | 6 | 28 | |
| | $t_{PLH}$ | | 85<br>86<br>87<br>88 | | | | | | | | | OUT | OUT | OUT | OUT | GND<br>GND<br>5.0 V<br>5.0 V | GND<br>5.0 V<br>GND<br>5.0 V | | | $\overline{E2}$ - 0<br>$\overline{E2}$ - 1<br>$\overline{E2}$ - 2<br>$\overline{E2}$ - 3 | | 25 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 89<br>90 | | | | | | | | | | OUT | OUT | | GND<br>5.0 V | IN | GND | | A0 - 1<br>A0 - 2 | 9<br>6 | 33<br>28 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 91<br>92 | | | | | | | | | | OUT | OUT | | GND<br>5.0 V | | | | A0 - 1<br>A0 - 2 | 9<br>6 | 33<br>25 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 93<br>94 | | | | | | | | | OUT | | | OUT | IN | GND<br>5.0 V | | | A1 - 0<br>A1 - 3 | 6<br>9 | 28<br>33 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 95<br>96 | | | | | | | | | OUT | | | OUT | | GND<br>5.0 V | | | A1 - 0<br>A1 - 3 | 6<br>9 | 25<br>33 | |

See notes at end of device type 01.





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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 $\overline{E1}$ | 2 A0 | 3 A1 | 4 $\overline{0}$ | 5 $\overline{1}$ | 6 $\overline{2}$ | 7 $\overline{3}$ | 8 GND | 9 $\overline{3}$ | 10 $\overline{2}$ | 11 $\overline{1}$ | 12 $\overline{0}$ | 13 A1 | 14 A0 | 15 $\overline{E2}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 10<br>$T_C$ = 125°C | $t_{PHL}$ | 3003 (Fig 5) | 97<br>98<br>99<br>100 | IN | GND<br>5.0 V<br>GND<br>5.0 V | GND<br>GND<br>5.0 V<br>5.0 V | OUT | OUT | OUT | OUT | GND | | | | | | | | 5.0 V | $\overline{E1}$ - $\overline{0}$<br>$\overline{E1}$ - $\overline{1}$<br>$\overline{E1}$ - $\overline{2}$<br>$\overline{E1}$ - $\overline{3}$ | 12 | 35 | ns |
| | $t_{PLH}$ | | 101<br>102<br>103<br>104 | | GND<br>5.0 V<br>GND<br>5.0 V | GND<br>GND<br>5.0 V<br>5.0 V | OUT | OUT | OUT | OUT | | | | | | | | | | $\overline{E1}$ - $\overline{0}$<br>$\overline{E1}$ - $\overline{1}$<br>$\overline{E1}$ - $\overline{2}$<br>$\overline{E1}$ - $\overline{3}$ | 6 | 39 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 105<br>106 | GND | IN | GND<br>5.0 V | | OUT | OUT | | | | | | | | | | | A0 - $\overline{1}$<br>A0 - $\overline{2}$ | 9<br>6 | 41<br>35 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 107<br>108 | | | GND<br>5.0 V | | OUT | OUT | | | | | | | | | | | A0 - $\overline{1}$<br>A0 - $\overline{2}$ | 9<br>6 | 50<br>39 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 109<br>110 | | GND<br>5.0 V | IN | OUT | | | OUT | | | | | | | | | | A1 - $\overline{0}$<br>A1 - $\overline{3}$ | 6<br>9 | 35<br>41 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 111<br>112 | | GND<br>5.0 V | | OUT | | | OUT | | | | | | | | | | A1 - $\overline{0}$<br>A1 - $\overline{3}$ | 6<br>9 | 39<br>50 | |
| | $t_{PHL}$ | | 113<br>114<br>115<br>116 | | | | | | | | | OUT | OUT | OUT | OUT | GND<br>GND<br>5.0 V<br>5.0 V | GND<br>5.0 V<br>GND<br>5.0 V | IN | | $\overline{E2}$ - $\overline{0}$<br>$\overline{E2}$ - $\overline{1}$<br>$\overline{E2}$ - $\overline{2}$<br>$\overline{E2}$ - $\overline{3}$ | 12 | 35 | |
| | $t_{PLH}$ | | 117<br>118<br>119<br>120 | | | | | | | | | OUT | OUT | OUT | OUT | GND<br>GND<br>5.0 V<br>5.0 V | GND<br>5.0 V<br>GND<br>5.0 V | | | $\overline{E2}$ - $\overline{0}$<br>$\overline{E2}$ - $\overline{1}$<br>$\overline{E2}$ - $\overline{2}$<br>$\overline{E2}$ - $\overline{3}$ | 6 | 39 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 121<br>122 | | | | | | | | | | OUT | OUT | | GND<br>5.0 V | IN | GND | | A0 - $\overline{1}$<br>A0 - $\overline{2}$ | 9<br>6 | 41<br>35 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 123<br>124 | | | | | | | | | | OUT | OUT | | GND<br>5.0 V | | | | A0 - $\overline{1}$<br>A0 - $\overline{2}$ | 9<br>6 | 50<br>39 | |
| | $t_{PHL}$<br>$t_{PHL}$ | | 125<br>126 | | | | | | | | | OUT | | | OUT | IN | GND<br>5.0 V | | | A1 - $\overline{0}$<br>A1 - $\overline{3}$ | 6<br>9 | 35<br>41 | |
| | $t_{PLH}$<br>$t_{PLH}$ | | 127<br>128 | | | | | | | | | OUT | | | OUT | | GND<br>5.0 V | | | A1 - $\overline{0}$<br>A1 - $\overline{3}$ | 6<br>9 | 39<br>50 | |
| 11 | Same tests, terminal conditions and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

NOTES:
1. X = Input may be high or low.
2. Output voltages shall be either:
   (a) H = 2.4 volts minimum and L = 0.4 volts maximum when using a high speed checker double comparator, or
   (b) H ≥ 1.5 volts and L < 1.5 volts when using a high speed checker single comparator.





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TABLE III. Group A inspection for device type 02

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Meas. terminal | Test limits Min | Test limits Max | Test limits Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | A1 | A2 | $\overline{LT}$ | $\overline{RBO}$ | $\overline{RBI}$ | A3 | A0 | GND | $\overline{e}$ | $\overline{d}$ | $\overline{c}$ | $\overline{b}$ | $\overline{a}$ | $\overline{g}$ | $\overline{f}$ | $V_{CC}$ | | | | |
| 1 $T_C$ = 25°C | $V_{OH}$ | 3007 | 1 | 0.8 V | 0.8 V | 2.0 V | -70 µA | 2.0 V | 0.8 V | 0.8 V | GND | | | | | | | | 4.5 V | $\overline{RBO}$ | 3.0 | | V |
| | $V_{OH}$ | 3007 | 2 | 2.0 V | 2.0 V | 0.8 V | -70 µA | 2.0 V | 2.0 V | 2.0 V | | | | | | | | | 4.5 V | $\overline{RBO}$ | 3.0 | | |
| | $V_{OL1}$ | 3006 | 3 | 0.8 V | 0.8 V | 2.0 V | 2.4 mA | 0.8 V | 0.8 V | 0.8 V | | | | | | | | | 4.5 V | $\overline{RBO}$ | | 0.4 | |
| | $V_{OL2}$ | 3006 | 4 | 0.8 V | 0.8 V | 2.0 V | 3.1 mA | 0 8 V | 0.8 V | 0.8 V | | | | | | | | | 5.5 V | $\overline{RBO}$ | | | |
| | $V_{OL3}$ | 3006 | 5 | | | 0.8 V | | | | | | 20 mA | | | | | | | 4.5 V | $\overline{e}$ | | | |
| | | | 6 | | | | | | | | | | 20 mA | | | | | | | $\overline{d}$ | | | |
| | | | 7 | | | | | | | | | | | 20 mA | | | | | | $\overline{c}$ | | | |
| | | | 8 | | | | | | | | | | | | 20 mA | | | | | $\overline{b}$ | | | |
| | | | 9 | | | | | | | | | | | | | 20 mA | | | | $\overline{a}$ | | | |
| | | | 10 | | | | | | | | | | | | | | 20 mA | | | $\overline{g}$ | | | |
| | | | 11 | | | | | | | | | | | | | | | 20 mA | | $\overline{f}$ | | | |
| | BVO | | 12 | 4.5 V | 4.5 V | 4.5 V | | | 4.5 V | 4.5 V | | 10 mA | | | | | | | 5.0 V | $\overline{e}$ | 30 | | |
| | | | 13 | | | | | | | | | | 10 mA | | | | | | | $\overline{d}$ | | | |
| | | | 14 | | | | | | | | | | | 10 mA | | | | | | $\overline{c}$ | | | |
| | | | 15 | | | | | | | | | | | | 10 mA | | | | | $\overline{b}$ | | | |
| | | | 16 | | | | | | | | | | | | | 10 mA | | | | $\overline{a}$ | | | |
| | | | 17 | | | | | | | | | | | | | | 10 mA | | | $\overline{g}$ | | | |
| | | | 18 | | | | | | | | | | | | | | | 10 mA | | $\overline{f}$ | | | |
| | $I_{IH1}$ | 3010 | 19 | 2.4 V | | GND | | | | | | | | | | | | | 5.5 V | A1 | 0 | 40 | µA |
| | $I_{IH1}$ | | 20 | | 2.4 V | GND | | | | | | | | | | | | | | A2 | | 40 | |
| | $I_{IH3}$ | | 21 | GND | GND | 2.4 V | | 5.5 V | GND | GND | | | | | | | | | | $\overline{LT}$ | | 200 | |
| | $I_{IH1}$ | | 22 | | | | | 2.4 V | | | | | | | | | | | | RBI | | 40 | |
| | | | 23 | | | GND | | | 2.4 V | | | | | | | | | | | A3 | | 40 | |
| | | | 24 | | | | | | | 2.4 V | | | | | | | | | | A0 | | 40 | |
| | $I_{IH2}$ | | 25 | 5.5 V | | | | | | | | | | | | | | | | A1 | | 100 | |
| | $I_{IH2}$ | | 26 | | 5.5 V | | | | | | | | | | | | | | | A2 | | 100 | |
| | $I_{IH4}$ | | 27 | GND | GND | 5.5 V | | 5 5 V | GND | GND | | | | | | | | | | $\overline{LT}$ | | 500 | |
| | $I_{IH2}$ | | 28 | | | | | 5.5 V | | | | | | | | | | | | RBI | | 100 | |
| | | | 29 | | | GND | | | 5.5 V | | | | | | | | | | | A3 | | 100 | |
| | | | 30 | | | GND | | | | 5.5 V | | | | | | | | | | A0 | | 100 | |
| | $I_{IL1}$ | 3009 | 31 | 0.4 V | | 4.5 V | | | | | | | | | | | | | | A1 | -0.5 | -1.4 | mA |
| | | | 32 | | 0.4 V | | | | | | | | | | | | | | | A2 | | | |
| | | | 33 | | | | | | 0.4 V | | | | | | | | | | | A3 | | | |
| | | | 34 | | | | | | | 0.4 V | | | | | | | | | | A0 | | | |
| | $I_{IL2}$ | | 35 | 4.5 V | 4.5 V | 0.4 V | | GND | GND | 4.5 V | | | | | | | | | | $\overline{LT}$ | -2 8 | -6.4 | |
| | $I_{IL3}$ | | 36 | | | | | 0.4 V | | | | | | | | | | | | RBI | -0.3 | -0 75 | |
| | $I_{OS}$ | 3011 | 37 | 4.5 V | 4.5 V | 4.5 V | GND | 4.5 V | GND | 4.5 V | | | | | | | | | 4.5 V | $\overline{RBO}$ | -0.64 | | |
| | $I_{OS}$ | 3011 | 38 | GND | GND | GND | GND | 4.5 V | 4.5 V | 4.5 V | | | | | | | | | 5.5 V | $\overline{RBO}$ | | -6.8 | |
| | $I_{CC}$ | 3005 | 39 | | | | | | | | | | | | | | | | 5.0 V | $V_{CC}$ | | 66 | |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = 125°C. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

See notes at end of device type 02.





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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 A1 | 2 A2 | 3 $\overline{LT}$ | 4 $\overline{RBO}$ | 5 $\overline{RBI}$ | 6 A3 | 7 A0 | 8 GND | 9 $\overline{e}$ | 10 $\overline{d}$ | 11 $\overline{c}$ | 12 $\overline{b}$ | 13 $\overline{a}$ | 14 $\overline{g}$ | 15 $\overline{f}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 7<br>$T_C$ = 25°C | Truth table test | | 40<br>41<br>42<br>43<br>44<br>45<br>46<br>47<br>48<br>49<br>50<br>51<br>52<br>53<br>54<br>55<br>56<br>57 | X<br>GND<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND<br>GND<br>5.0 V<br>5.0 V | X<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V | GND<br>5.0 V | H<br>L<br>H<br>L | X<br>GND<br>5.0 V<br>X | X<br>GND<br>5.0 V | X<br>GND<br>GND<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>5.0 V | GND | L H L L L H H H L L H L H L H L<br>H H L L H H L L H H L L H H L L<br>L L L H H H L L L L H H H H L L<br>H L H L H L H L H L H L H L H L | L H L H L H L H L H L H L H L H<br>L L H H L L H H L L H H L L H H<br>L L L L H H H H L L L L H H H H | L H L H H L<br>H | L H L H L L H H L L H H L L H H | L H L H L H L H L H L H L H L H | L H H H L<br>H L L H | L H L L H H L L H H L L H H L L | 5.0 V | | See notes 1 and 2 | | |
| 8 | Truth table test | | 58 thru 75 | Same tests as subgroup 7, except $T_C$ = 125°C. | | | | | | | | | | | | | | | | | | | |
| | Truth table test | | 76 thru 93 | Same tests as subgroup 7, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | |
| 9<br>$T_C$ = 25°C | $t_{PHL}$ | 3003 (Fig 5) | 94<br>95<br>96<br>97<br>98<br>99<br>100<br>101 | GND<br>5.0 V<br>5.0 V<br>GND | GND<br>5.0 V<br>GND<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND | 5.0 V | OUT | 5.0 V<br>GND | GND<br>5.0 V<br>GND | IN | GND | OUT | OUT | OUT | OUT | OUT | OUT | OUT | 5.0 V | A0 - $\overline{a}$<br>A0 - $\overline{b}$<br>A0 - $\overline{c}$<br>A0 - $\overline{d}$<br>A0 - $\overline{e}$<br>A0 - $\overline{f}$<br>A0 - $\overline{g}$<br>A0 - $\overline{RBO}$ | 50<br>10 | 700<br>400 | ns |
| | $t_{PLH}$ | | 102<br>103<br>104<br>105<br>106<br>107<br>108<br>109 | GND<br>5.0 V<br>5.0 V<br>GND | GND<br>5.0 V<br>GND<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND | | OUT | 5.0 V<br>GND | 5.0 V<br>GND | | | OUT | OUT | OUT | OUT | OUT | OUT | OUT | | A0 - $\overline{a}$<br>A0 - $\overline{b}$<br>A0 - $\overline{c}$<br>A0 - $\overline{d}$<br>A0 - $\overline{e}$<br>A0 - $\overline{f}$<br>A0 - $\overline{g}$<br>A0 - $\overline{RBO}$ | 30<br>60 | 500<br>700 | |
| | $t_{PHL}$ | | 110<br>111<br>112<br>113<br>114<br>115<br>116 | IN | GND<br>5.0 V<br>GND<br>5.0 V<br>GND<br>GND<br>GND | | OUT | 5.0 V | 5.0 V | 5.0 V<br>GND<br>GND<br>5.0 V<br>5.0 V<br>GND<br>GND | | OUT | OUT | OUT | OUT | OUT | | OUT | | A1 - $\overline{a}$<br>A1 - $\overline{b}$<br>A1 - $\overline{c}$<br>A1 - $\overline{d}$<br>A1 - $\overline{e}$<br>A1 - $\overline{f}$<br>A1 - $\overline{RBO}$ | 50<br>10 | 400 | |

See notes at end of device type 02.





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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high $\geq$ 2.0 V; or low $\leq$ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 A1 | 2 A2 | 3 $\overline{LT}$ | 4 $\overline{RBO}$ | 5 $\overline{RBI}$ | 6 A3 | 7 A0 | 8 GND | 9 $\bar{e}$ | 10 $\bar{d}$ | 11 $\bar{c}$ | 12 $\bar{b}$ | 13 $\bar{a}$ | 14 $\bar{g}$ | 15 $\bar{f}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9 $T_C$ = 25°C | $t_{PLH}$ | 3003 (Fig 5) | 117 | IN | GND | 5.0 V | | 5.0 V | GND | 5.0 V | GND | | | | | OUT | | | 5.0 V | A1 - $\bar{a}$ | 30 | 500 | ns |
| | | | 118 | | 5.0 V | | | | | GND | | | | | OUT | | | | | A1 - $\bar{b}$ | | | |
| | | | 119 | | GND | | | | | GND | | | | OUT | | | | | | | A1 - $\bar{c}$ | | | |
| | | | 120 | | 5.0 V | | | | | 5.0 V | | | OUT | | | | | | | | A1 - $\bar{d}$ | | | |
| | | | 121 | | GND | | | | | 5.0 V | | OUT | | | | | | | | | A1 - $\bar{e}$ | | | |
| | | | 122 | | GND | | | | | GND | | | | | | | | | | | A1 - $\bar{f}$ | | | |
| | | | 123 | | GND | | OUT | | 5.0 V | GND | | | | | | | | OUT | | | A1 - $\overline{RBO}$ | 60 | 700 | |
| | $t_{PHL}$ | | 124 | GND | IN | | | | GND | GND | | | | | | OUT | | | | | A2 - $\bar{a}$ | 50 | | |
| | | | 125 | | | | | | | 5.0 V | | | | OUT | | | | | | | A2 - $\bar{c}$ | | | |
| | | | 126 | | | | | | | GND | | | OUT | | | | | | | | A2 - $\bar{d}$ | | | |
| | | | 127 | | | | | | | 5.0 V | | OUT | | | | | | | | | A2 - $\bar{e}$ | | | |
| | $t_{PLH}$ | | 128 | | | | | | | GND | | | | | | OUT | | | | | A2 - $\bar{a}$ | 30 | 500 | |
| | | | 129 | | | | | | | 5.0 V | | | | OUT | | | | | | | A2 - $\bar{c}$ | | | |
| | | | 130 | | | | | | | GND | | | OUT | | | | | | | | A2 - $\bar{d}$ | | | |
| | | | 131 | | | | | | | 5.0 V | | OUT | | | | | | | | | A2 - $\bar{e}$ | | | |
| | $t_{PHL}$ | | 132 | | GND | | | | IN | 5.0 V | | | | | OUT | | | | | | A3 - $\bar{b}$ | 50 | 700 | |
| | $t_{PHL}$ | | 133 | 5.0 V | 5.0 V | | OUT | | | GND | | | | | | | | | | | A3 - $\overline{RBO}$ | 10 | 400 | |
| | $t_{PLH}$ | | 134 | GND | GND | | | | | 5.0 V | | | | | OUT | | | | | | A3 - $\bar{b}$ | 30 | 500 | |
| | $t_{PLH}$ | | 135 | 5.0 V | 5.0 V | | OUT | | | GND | | | | | | | | | | | A3 - $\overline{RBO}$ | 60 | 700 | |
| | $t_{PHL}$ | | 136 | GND | GND | | | IN | GND | | | | | | | OUT | | | | | $\overline{RBI}$ - $\bar{a}$ | 30 | 750 | |
| | $t_{PHL}$ | | 137 | | | | OUT | | | | | | | | | | | | | | $\overline{RBI}$-$\overline{RBO}$ | 10 | 175 | |
| | $t_{PLH}$ | | 138 | | | | | | | | | | | | | OUT | | | | | $\overline{RBI}$ - $\bar{a}$ | 30 | 500 | |
| | $t_{PLH}$ | | 139 | | | | OUT | | | | | | | | | | | | | | $\overline{RBI}$-$\overline{RBO}$ | 40 | 450 | |
| | $t_{PHL}$ | | 140 | | | IN | | GND | | | | | | | OUT | | | | | | $\overline{LT}$ - $\bar{b}$ | 50 | 700 | |
| | $t_{PHL}$ | | 141 | | | | | | | | | | | | | | OUT | | | | $\overline{LT}$ - $\bar{g}$ | 50 | 700 | |
| | $t_{PHL}$ | | 142 | | | | OUT | | | | | | | | | | | | | | $\overline{LT}$ -$\overline{RBO}$ | 10 | 350 | |
| | $t_{PLH}$ | | 143 | | | | | | | | | | | | OUT | | | | | | $\overline{LT}$ - $\bar{b}$ | 30 | 500 | |
| | $t_{PLH}$ | | 144 | | | | | | | | | | | | | | OUT | | | | $\overline{LT}$ - $\bar{g}$ | | 500 | |
| | $t_{PLH}$ | | 145 | | | | OUT | | | | | | | | | | | | | | $\overline{LT}$ -$\overline{RBO}$ | | 350 | |
| | $t_{PHL}$ | | 146 | | | 5.0 V | IN | 5.0 V | | | | | | OUT | | | | | | | $\overline{RBO}$ - $\bar{c}$ | | 400 | |
| | $t_{PLH}$ | | 147 | | | 5.0 V | IN | 5.0 V | | | | | | OUT | | | | | | | $\overline{RBO}$ - $\bar{c}$ | | 300 | |
| 10 $T_C$ = 125 C | $t_{PHL}$ | 3003 (Fig 5) | 148 | | GND | 5.0 V | | 5.0 V | GND | IN | GND | | | | OUT | OUT | | | 5.0 V | A0 - $\bar{a}$ | 50 | 1,200 | |
| | | | 149 | | 5.0 V | | | | | | | | | OUT | | | | | | | A0 - $\bar{b}$ | | | |
| | | | 150 | | GND | | | | | | | | OUT | | | | | | | | A0 - $\bar{c}$ | | | |
| | | | 151 | | GND | | | | | | | | | | | | | | | | A0 - $\bar{d}$ | | | |
| | | | 152 | | GND | | | | 5.0 V | | | OUT | | | | | | | | | A0 - $\bar{e}$ | | | |
| | | | 153 | 5.0 V | 5.0 V | | | | GND | | | | | | | | | OUT | | | A0 - $\bar{f}$ | | | |
| | | | 154 | 5.0 V | 5.0 V | | | | | | | | | | | | OUT | | | | A0 - $\bar{g}$ | | | |
| | | | 155 | GND | GND | | OUT | GND | | | | | | | | | | | | | A0 - $\overline{RBO}$ | 10 | 500 | |
| | $t_{PLH}$ | | 156 | | GND | | | 5.0 V | | | | | | | | OUT | | | | | A0 - $\bar{a}$ | 30 | 1,100 | |
| | | | 157 | | 5.0 V | | | | | | | | | OUT | OUT | | | | | | A0 - $\bar{b}$ | | | |
| | | | 158 | | GND | | | | | | | | OUT | | | | | | | | A0 - $\bar{c}$ | | | |
| | | | 159 | | GND | | | | | | | | | | | | | | | | A0 - $\bar{d}$ | | | |
| | | | 160 | | GND | | | | 5.0 V | | | OUT | | | | | | | | | A0 - $\bar{e}$ | | | |
| | | | 161 | 5.0 V | 5.0 V | | | | GND | | | | | | | | | OUT | | | A0 - $\bar{f}$ | | | |
| | | | 162 | 5.0 V | 5.0 V | | | | GND | | | | | | | | OUT | | | | A0 - $\bar{g}$ | | | |
| | | | 163 | GND | GND | | OUT | GND | GND | | | | | | | | | | | | A0 - $\overline{RBO}$ | 60 | 1,100 | |

See notes at end of device type 02.



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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open)

| Subgroup | Symbol | MIL-STD-883 method | Case E, F Test No. | 1 A1 | 2 A2 | 3 $\overline{LT}$ | 4 $\overline{RBO}$ | 5 $\overline{RBI}$ | 6 A3 | 7 A0 | 8 GND | 9 $\bar{e}$ | 10 $\bar{d}$ | 11 $\bar{c}$ | 12 $\bar{b}$ | 13 $\bar{a}$ | 14 $\bar{g}$ | 15 $\bar{f}$ | 16 $V_{CC}$ | Meas. terminal | Test limits Min | Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 10 $T_C$ 125°C | $t_{PHL}$ | 3003 (Fig 5) | 164 | IN | GND | 5.0 V | | 5.0 V | GND | 5.0 V | GND | | | | OUT | OUT | | | 5.0 V | A1 - $\bar{a}$ | 50 | 1,200 | ns |
| | | | 165 | | 5.0 V | | | | | GND | | | | | | | | | | A1 - $\bar{b}$ | | | |
| | | | 166 | | GND | | | | | GND | | | | OUT | | | | | | A1 - $\bar{c}$ | | | |
| | | | 167 | | 5.0 V | | | | | 5.0 V | | OUT | | | | | | | | A1 - $\bar{d}$ | | | |
| | | | 168 | | GND | | | | | 5.0 V | | | OUT | | | | | | | A1 - $\bar{e}$ | | | |
| | | | 169 | | | | | | | GND | | | | | | | | OUT | | A1 - $\bar{f}$ | | | |
| | | | 170 | | | | OUT | | 5.0 V | GND | | | | | | | | | | A1 - $\overline{RBO}$ | 10 | 500 | |
| | $t_{PLH}$ | | 171 | | 5.0 V | | | | GND | 5.0 V | | | | | OUT | OUT | | | | A1 - $\bar{a}$ | 30 | 1,100 | |
| | | | 172 | | GND | | | | | GND | | | | OUT | | | | | | A1 - $\bar{b}$ | | | |
| | | | 173 | | 5.0 V | | | | | GND | | OUT | | | | | | | | A1 - $\bar{c}$ | | | |
| | | | 174 | | GND | | | | | 5.0 V | | | OUT | | | | | | | A1 - $\bar{d}$ | | | |
| | | | 175 | | GND | | | | | 5.0 V | | | | | | | | OUT | | A1 - $\bar{e}$ | | | |
| | | | 176 | | GND | | | | | GND | | | | | | | | | | A1 - $\bar{f}$ | | | |
| | | | 177 | | | | OUT | | 5.0 V | GND | | | | | | | | | | A1 - $\overline{RBO}$ | 60 | 1,100 | |
| | $t_{PHL}$ | | 178 | GND | IN | | | | GND | GND | | | | | | OUT | | | | A2 - $\bar{a}$ | 50 | 1,200 | |
| | | | 179 | | | | | | | 5.0 V | | | | | | | | | | A2 - $\bar{c}$ | | | |
| | | | 180 | | | | | | | GND | | | OUT | | | | | | | A2 - $\bar{d}$ | | | |
| | | | 181 | | | | | | | 5.0 V | | OUT | | | | | | | | A2 - $\bar{e}$ | | | |
| | $t_{PLH}$ | | 182 | | | | | | | GND | | | | | | OUT | | | | A2 - $\bar{a}$ | 30 | 1,100 | |
| | | | 183 | | | | | | | 5.0 V | | | | | | | | | | A2 - $\bar{c}$ | | | |
| | | | 184 | | | | | | | GND | | | OUT | | | | | | | A2 - $\bar{d}$ | | | |
| | | | 185 | | | | | | | 5.0 V | | OUT | | | | | | | | A2 - $\bar{e}$ | | | |
| | $t_{PHL}$ | | 186 | | GND | | | | IN | 5.0 V | | | | | OUT | | | | | A3 - $\bar{b}$ | 50 | 1,200 | |
| | $t_{PHL}$ | | 187 | 5.0 V | 5.0 V | | OUT | | | GND | | | | | | | | | | A3 - $\overline{RBO}$ | 10 | 500 | |
| | $t_{PLH}$ | | 188 | GND | GND | | | | | 5.0 V | | | | | OUT | | | | | A3 - $\bar{b}$ | 30 | 1,100 | |
| | $t_{PLH}$ | | 189 | 5.0 V | 5.0 V | | OUT | | | GND | | | | | | | | | | A3 - $\overline{RBO}$ | 60 | 1,100 | |
| | $t_{PHL}$ | | 190 | GND | GND | | | IN | GND | | | | | | | OUT | | | | $\overline{RBI}$ - $\bar{a}$ | 30 | 1,300 | |
| | $t_{PHL}$ | | 191 | | | | OUT | | | | | | | | | | | | | $\overline{RBI}$-$\overline{RBO}$ | 10 | 400 | |
| | $t_{PLH}$ | | 192 | | | | | | | | | | | | | OUT | | | | $\overline{RBI}$ - a | 30 | 1,200 | |
| | $t_{PLH}$ | | 193 | | | | OUT | | | | | | | | | | | | | $\overline{RBI}$-$\overline{RBO}$ | 40 | 700 | |
| | $t_{PHL}$ | | 194 | | | IN | | GND | | | | | | | OUT | | OUT | | | $\overline{LT}$ - $\bar{b}$ | 50 | 1,200 | |
| | $t_{PHL}$ | | 195 | | | | | | | | | | | | | | | | | $\overline{LT}$ - $\bar{g}$ | 50 | 1,200 | |
| | $t_{PHL}$ | | 196 | | | | OUT | | | | | | | | | | | | | $\overline{LT}$ - $\overline{RBO}$ | 10 | 80 | |
| | $t_{PLH}$ | | 197 | | | | | | | | | | | | OUT | | OUT | | | $\overline{LT}$ - $\bar{b}$ | 30 | 1,100 | |
| | $t_{PLH}$ | | 198 | | | | | | | | | | | | | | | | | $\overline{LT}$ - $\bar{g}$ | | 1,100 | |
| | $t_{PLH}$ | | 199 | | | | OUT | | | | | | | | | | | | | $\overline{LT}$ - $\overline{RBO}$ | | 700 | |
| | $t_{PHL}$ | | 200 | | | 5.0 V | IN | 5.0 V | | | | | | OUT | | | | | | $\overline{RBO}$ - $\bar{c}$ | | 600 | |
| | $t_{PLH}$ | | 201 | | | 5.0 V | IN | 5.0 V | | | | | | OUT | | | | | | $\overline{RBO}$ - $\bar{c}$ | | 700 | |
| 11 | Same tests, terminal conditions and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

NOTES:
1 X = Input may be high or low.
2 Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volts maximum when using a high speed checker double comparator, or
(b) H ≥ 1.5 volts and L < 1.5 volts when using a high speed checker single comparator.





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6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

| | |
|---|---|
| GND | Electrical ground (common terminal). |
| $I_{IN}$ | Current flowing into an input terminal. |
| $T_C$ | Case temperature. |
| $V_{IN}$ | Voltage level at an input terminal. |

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

| Military device type | Generic-industry type |
|---|---|
| 01 | 9321 |
| 02 | 9317 |

6.8 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.



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Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-38510/158A | MICROCIRCUITS, DIGITAL, TTL, DECODERS, MONOLITHIC SILICON |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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